

A DISTRIBUTION-FREE METHOD OF ANALIZING A 2^m FACTORIAL
EXPERIMENT*

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1. Summary and Introduction

The major problem in distribution-free analysis of variance methods is the testing of effects independently of each other. In this paper a distribution-free method is suggested for testing any effect independently of other effects in a 2^m factorial experiment with an equal number of observations per cell. The rationale of the method is to reduce the problem of testing for any effect to a two-sample problem of location and then to apply any of the existing distribution-free two-sample tests. For the case of all interactions of order p and higher being zero an alternative test is suggested which is based on the sum of 2^{p-1} independent two-sample test statistics. These procedures are illustrated with Wilcoxon's (1945) two-sample test statistic and in the case of normality their asymptotic relative efficiencies relative to the parametric method are shown to be 3/π for location alternatives. Finally sufficient small sample probability tables of the sum of independent Wilcoxon test statistics are given.

2. The 2² experiment

Consider a 2² experiment. Let X_{hij} be the j th observation in random order in cell i of row h for $h=1,2$; $i=1,2$; $j=1,\dots,n$. The fixed effects model is assumed to hold, namely

$$X_{hij} = R_h + C_i + (RC)_{hi} + Z_{hij}$$

where $R_1 = -R_2 = R$, $C_1 = -C_2 = C$, $(RC)_{11} = (RC)_{22} = -(RC)_{12} = -(RC)_{21} = (RC)$, and the Z 's are independent observations of a random variable with distribution function F . It is assumed that $F(x)$ and $f(x) = d/dx F(x)$ are everywhere continuous.

2.1 Testing of interaction

The hypothesis of no interaction is

$$(2.1) \quad (RC) = 0$$

and this has to be tested, independently of row and column effects, against the alternative

$$(2.2) \quad (RC) \neq 0$$

Under (2.1) the sets $\{Y_{1j} = X_{11j} + X_{22j}\}$ and $\{Y_{2j} = X_{12j} + X_{21j}\}$, $j=1,\dots,n$, can be regarded as two random and independent sam-

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ples from identical populations with continuous distribution function $F_1(y)$, (F_1 denotes the distribution function of the sum of 2 independent identically distributed random variables each with distribution function F) while under (2.2) these sets can be regarded as random and independent samples from populations with continuous distribution functions $F_1\{y-2(RC)\}$ and $F_1\{y+2(RC)\}$ respectively. Herewith column and row effects are eliminated and the problem reduces to a two-sample problem of location so that any of the existing two-sample tests can be applied. As an illustration consider Wilcoxon's test, i.e.

$$V_1 = \sum_{j_1=1}^n \sum_{j_2=1}^n v(y_{1j_1}, y_{2j_2}),$$

$$v(a,b) = \begin{cases} \frac{1}{2} & \text{for } a > b \\ -\frac{1}{2} & \text{for } a < b. \end{cases}$$

Under a near alternative of the form

$$(2.3) \quad (RC) = \frac{\alpha}{\sqrt{n}}$$

it is well known that $12V_1^2/n^2(2n+1)$ has in the limit as n tends to infinity a non-central χ^2 distribution with one degree of freedom and non-centrality parameter

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{12}{n^2(2n+1)} E^2\{V_1 | (RC) = \frac{\alpha}{\sqrt{n}}\} \\ &= \lim_{n \rightarrow \infty} 6n \left[\int \{F_1(y + \frac{2\alpha}{\sqrt{n}}) - F_1(y - \frac{2\alpha}{\sqrt{n}})\} dF_1(y - \frac{2\alpha}{\sqrt{n}}) \right]^2 \end{aligned}$$

and under the assumption of continuity this reduces to

$$96 \alpha^2 \{f_1^2(y) dy\}^2$$

$$\text{where } f_1(y) = \frac{d}{dy} F_1(y).$$

The parametric test for this problem is based on

$$\frac{n \sum_h \sum_i (X_{hi.} - X_{h..} - X_{.i.} + X_{...})^2}{\frac{1}{4(n-1)} \sum_h \sum_i (X_{hij} - X_{hi.})^2}$$

where $X_{hi.}$, $X_{h..}$, $X_{.i.}$ and $X_{...}$ are the cell, row, column and total means. Under (2.3) this statistic has in the limit as n tends to infinity a non-central χ^2 distribution with one degree of freedom and non-centrality parameter $4\alpha^2/\sigma^2$ where σ^2 is the variance of the random variable with distribution function F . The asymptotic relative efficiency of the test based on V relative to the parametric test is thus

$$(2.4) \quad 24 \sigma^2 \{f_1^2(y) dy\}^2$$

$$\text{or} \quad 12 \sigma_{F_1}^2 \{f_1^2(y) dy\}^2$$

where $\sigma_{F_1}^2 = 2\sigma^2$ is the variance of the random variable with distribution function F_1 . Specifically if F is a normal distribution function (2.4) is equal to $3/\pi$ which is the same as the

a.r.e. of Wilcoxon's test relative to the t-test. Similarly, cf. Hodges & Lehmann (1956), the minimum value of (2.4) is 108/125. The above correspondence, however, is not complete because for the rectangular distribution (2.4) is 8/9 as compared with unity for Wilcoxon's test relative to the t-test.

2.2 Testing of main effects

Consider row effects. The hypothesis to be tested is

$$(2.5) \quad R = 0$$

against

$$(2.6) \quad R \neq 0$$

Under (2.6) the sets $\{Y_{1j}=X_{11j}+X_{12j}\}$ and $\{Y_{2j}=X_{21j}+X_{22j}\}$, $j=1, \dots, n$, can again be regarded as two random and independent samples from identical populations with continuous distribution function $F_1(y)$, while under (2.6) as random and independent samples from populations with continuous distribution functions $F_1(y-2R)$ and $F_1(y+2R)$ respectively, and the same procedure as above applies.

If it is known that interaction is not present, i.e.

$$R = 0, (RC) = 0$$

must be tested against

$$R \neq 0, (RC) = 0$$

then, cf. Crouse (1967), the sum of the two two-sample test statistics applied to $\{X_{1jk}\}$ and $\{X_{2jk}\}$, $k=1, \dots, n$, for $j=1, 2$, can also be used as test statistic. Since this procedure leads to a test statistic which is for the same n less discrete than the above procedure it is preferred, especially for small n .

3. The 2^m experiment

From the 2² experiment the following generalization to the 2^m experiment readily follows.

The observations are denoted by $X_{i_1 i_2 \dots i_m k}$, $i_j=1, 2$; $j=1, \dots, m$; $k=1, \dots, n$, and it is assumed that the n observations in every cell are in a random order. The appropriate fixed effects model is

$$(3.1) \quad X_{i_1 i_2 \dots i_m k} = \sum_{j=1}^m (-1)^{\delta_{ij}} I_j + \sum_{1 \leq j_1 < j_2 \leq m} (-1)^{\delta_{i j_1 j_2}} I_{j_1 j_2} + \dots + (-1)^{\delta_{i i_1 i_2 \dots i_m}} I_{1 \dots m} + Z_{i_1 i_2 \dots i_m k}.$$

where the I 's denote zero order, first order, ..., m th order interaction, and

$$\delta_{i_1 i_2 \dots i_p} = \begin{cases} 1 & \text{if an odd number of the set } \{i_1 i_2 \dots i_p\} \text{ is equal to two} \\ 0 & \text{if an even number of the set } \{i_1 i_2 \dots i_p\} \text{ is equal to two} \end{cases}$$

for $p=1, 2, \dots, m$, and the Z 's are random and independent obser-

variations of a random variable with continuous distribution function F .

The general problem is to test the hypothesis $I_{j_1 j_2 \dots j_p} = 0$ against $I_{j_1 j_2 \dots j_p} \neq 0$. Without loss of generality it can be assumed that $j_i = i$, $i=1, \dots, p$, and the problem is to test

$$(3.2) \quad I_{12 \dots p} = 0$$

against

$$(3.3) \quad I_{12 \dots p} \neq 0$$

independently of the other effects. Let

$$Y_{1k} = \sum_{i_1 i_2 \dots i_m} (1 - \delta_{i_1 i_2 \dots i_p}) X_{i_1 i_2 \dots i_m k}$$

$$Y_{2k} = \sum_{i_1 i_2 \dots i_m} \delta_{i_1 i_2 \dots i_p} X_{i_1 i_2 \dots i_m k}$$

for $k=1, \dots, n$, and consider

$$\Delta = \sum_{i_1 i_2 \dots i_m} (1 - \delta_{i_1 i_2 \dots i_p}) (-1)^{\delta_{i_1 i_2 \dots i_p}} I_{j_1 j_2 \dots j_q}$$

If a subscript j_h , say j_1 , is not contained in $\{1, \dots, p\}$ then $\Delta=0$ because

$$(1 - \delta_{i_1 i_2 \dots i_p}) (-1)^{\delta_{i_1 i_2 \dots i_p}} = -(1 - \delta_{i_1 i_2 \dots i_p}) \cdot (-1)^{\delta_{i_1 i_2 \dots i_p}}$$

for all $i_1, i_2, \dots, i_m$, where $\bar{i}_h$ is the complement of i_h , i.e. if $i_h=1$ then $\bar{i}_h=2$ and conversely.

On the other hand if a subscript i_h , say i_1 , is not contained in $\{j_1, j_2, \dots, j_q\}$ while the latter is a subset of $\{1, 2, \dots, p\}$, say with $j_1=2$, then $\Delta=0$ because

$$(1 - \delta_{i_1 i_2 i_3 \dots i_p}) (-1)^{\delta_{i_1 i_2 i_3 \dots i_p}} = -(1 - \delta_{i_1 i_2 i_3 \dots i_p}) \cdot (-1)^{\delta_{i_1 i_2 i_3 \dots i_p}}$$

for all $i_1, i_2, \dots, i_m$. It follows that the only non-vanishing Δ is

$$\begin{aligned} & \sum_{i_1 i_2 \dots i_m} (1 - \delta_{i_1 i_2 \dots i_p}) (-1)^{\delta_{i_1 i_2 \dots i_p}} I_{12 \dots p} \\ & = 2^{m-1} I_{12 \dots p}. \end{aligned}$$

Substitution of (3.1) into Y_{1k} now readily reduces this to

$$Y_{1k} = 2^{m-1} I_{12 \dots p} + U_{1k}$$

and similarly it can be shown that

$$Y_{2k} = -2^{m-1} I_{12 \dots p} + U_{2k},$$

for $k=1, \dots, n$, where the U 's can be regarded as random and independent observations of a random variable with continuous distribution function $F_{m-1}(u)$ (F_{m-1} denotes the distribution function of 2^{m-1} independent identically distributed random

variables each with distribution function F). The problem of testing (3.2) against (3.3) thus reduces to a two-sample problem of location, and application of Wilcoxon's test results, as in Section 2, in an a.r.e. of

$$12 \sigma_{F_{m-1}}^2 \{ \int f_{m-1}^2(y) dy \}^2$$

relative to the parametric test where $f_{m-1}(y) = d/dy F_{m-1}(y)$ and $\sigma_{F_{m-1}}^2 = 2^{m-1} \sigma^2$ is the variance of the random variable with distribution function F_{m-1} .

If it is known that all interactions of order p and higher are zero then any of the hypotheses concerning interaction of order less than p , i.e.

$$I_{12\dots p'} = 0 \text{ for } p' \leq p$$

can also be tested against

$$I_{12\dots p'} \neq 0$$

by an alternative method. Let

$$Y_{1i_{p+1}\dots i_{m-k}} = \sum_{i_1\dots i_p} (1 - \delta_{i_1\dots i_p}) X_{i_1\dots i_{m-k}}$$

$$Y_{2i_{p+1}\dots i_{m-k}} = \sum_{i_1\dots i_p} \delta_{i_1\dots i_p} X_{i_1\dots i_{m-k}}$$

where $i_{p+r}=1,2$; $r=1,\dots,(m-p)$; $k=1,\dots,n$, and consider

$$\Delta = \sum_{i_1\dots i_p} (1 - \delta_{i_1\dots i_p}) (-1)^{\delta_{i_1\dots i_p}} I_{j_1\dots j_q}$$

As above $\Delta=0$ if there exists a subscript j_h for which $p' < j_h \leq p$ or if a subscript $i_r, r \leq p'$, is not contained in $\{j_1, \dots, j_q\}$ while the latter is a subset of $\{1, \dots, p'\}$. It follows that the only non-vanishing Δ 's are, firstly

$$\sum_{i_1\dots i_p} (1 - \delta_{i_1\dots i_p}) (-1)^{\delta_{i_1\dots i_p}} I_{1\dots p'} = 2^{p-1} I_{1\dots p'}$$

and secondly when none of the j_h 's are numbers of $\{1, \dots, p\}$, in which case Δ becomes,

$$2^{p-1} (-1)^{\delta_{i_1\dots i_p}} I_{j_1\dots j_q}$$

Substitution of (3.1), under the assumption that interactions of order p and higher are zero, in $Y_{1i_{p+1}\dots i_{m-k}}$ readily reduces this to

$$\begin{aligned} Y_{1i_{p+1}\dots i_{m-k}} &= 2^{p-1} I_{1\dots p'} \\ &+ 2^{p-1} \left\{ \sum_{j=p+1}^m (-1)^{\delta_{ij}} I_j + \sum_{p < j_1 < j_2 \leq m} (-1)^{\delta_{ij_1 j_2}} I_{j_1 j_2} + \dots \right. \\ &\left. + (-1)^{\delta_{ip+1\dots i_m}} I_{p+1\dots m} \right\} + U_{1i_{p+1}\dots i_{m-k}} \end{aligned}$$

i.e. $Y_{1i_{p+1}\dots i_{m-k}} = 2^{p-1} I_{1\dots p'} + U_{1i_{p+1}\dots i_{m-k}}$

and similarly it can be shown that

$$Y_{2i_{p+1} \dots i_m k} = -2^{p-1} I_{1 \dots p}^{t+u+U} 2i_{p+1} \dots i_m k$$

for $i_{p+r}=1,2; r=1,\dots,(m-p); k=1,\dots,n$ and where the U 's can be regarded as random and independent observations of a random variable with continuous distribution function $F_{p-1}(u)$. For every combination of $i_{p+1}, \dots, i_m$ the testing of (3.2) against (3.3) reduces to a two-sample problem of location and if $T_{i_{p+1} \dots i_m}$ denotes the test corresponding to $i_{p+1}, \dots, i_m$ the combined test becomes

$$\sum_{i_{p+1} \dots i_m} T_{i_{p+1} \dots i_m}$$

In the case of the Wilcoxon test the latter is the sum of 2^{m-p} independent Wilcoxon statistics and the asymptotic relative efficiency of this test relative to the parametric test can again be shown to be

$$12 \sigma_{F_{p-1}}^2 \{ \int f_{F_{p-1}}^2(y) dy \}^2$$

where $d/dy F_{p-1}(y) = f_{p-1}(y)$ and $\sigma_{F_{p-1}}^2$ is the variance of the random variable with distribution function F_{p-1} .

4. Practical considerations

When applying the above methods the following points are of practical importance.

(a) These methods are applicable to a fixed effects 2^m design with an equal number of observations per cell.

(b) Due to the discreteness of the test statistics employed, cf. Owen (1962) for the probability tables of the Wilcoxon test statistic, there must in general be at least four observations per cell to test at the 5% level of significance. This restriction is somewhat relieved when higher order interactions can be assumed to be zero. Cf. Tables 1 and 2.

(c) The observations in each cell are assumed to be in a random order. If this assumption is not realistic the observations in each cell must be ordered by a random device.

(d) Sufficient probability tables exist for the Wilcoxon test statistic while Tables 1 and 2 give the small sample probabilities of the sum of two and four independent Wilcoxon test statistics respectively. From Table 1 it is clear that for the sum of two Wilcoxon test statistics the normal approximation with continuity correction can safely be used for $n > 4$, the maximum difference being 0.005 for $n=4$, while from Table 2 the normal approximation with continuity correction can be used for the sum of four or more Wilcoxon statistics even for $n=2$.

TABLE 1

Cumulative probability distribution of the sum U of two independent two-sample Wilcoxon test statistics

		u	P(U <u>≤</u> u)			u	P(U <u>≤</u> u)
n=2	0	0.028			5	0.778	
	1	0.083			6	0.917	
	2	0.222			7	0.972	
	3	0.389			8	1.000	
	4	0.611					
n=3	0	0.003			10	0.673	
	1	0.008			11	0.773	
	2	0.020			12	0.855	
	3	0.045			13	0.915	
	4	0.085			14	0.955	
	5	0.145			15	0.980	
	6	0.228			16	0.993	
	7	0.328			17	0.998	
	8	0.440			18	1.000	
	9	0.560					
		Normal approximation*)				Normal approximation*)	
u	P(U <u>≤</u> u)	p'	p''	u	P(U <u>≤</u> u)	p'	p''
0	0.000	0.000	0.001	17	0.617	0.579	0.622
1	0.001	0.001	0.002	18	0.691	0.659	0.695
2	0.002	0.002	0.003	19	0.757	0.729	0.761
3	0.004	0.004	0.005	20	0.816	0.794	0.821
4	0.008	0.007	0.010	21	0.865	0.846	0.869
5	0.014	0.013	0.016	22	0.905	0.889	0.908
6	0.025	0.021	0.026	23	0.936	0.924	0.937
7	0.041	0.034	0.042	24	0.959	0.949	0.958
8	0.064	0.052	0.063	25	0.975	0.966	0.974
9	0.095	0.076	0.092	26	0.986	0.979	0.984
10	0.135	0.111	0.131	27	0.992	0.988	0.990
11	0.184	0.154	0.179	28	0.996	0.993	0.995
12	0.243	0.206	0.239	29	0.998	0.996	0.997
13	0.309	0.271	0.305	30	0.999	0.998	0.999
14	0.383	0.341	0.378	31	1.000	0.999	0.999
15	0.460	0.421	0.460	32	1.000	1.000	1.000
16	0.540	0.500	0.540				

*) p' is without and p'' with a continuity correction.

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TABLE 2

Cumulative probability distribution of the sum V of four independent two-sample Wilcoxon test statistics

Normal approximation				Normal approximation				
n=2	v	P(V≤v)	p'	p''	v	P(V≤v)	p'	p''
	0	0.001	0.001	0.002	9	0.715	0.652	0.719
	1	0.004	0.003	0.006	10	0.830	0.782	0.834
	2	0.015	0.010	0.017	11	0.910	0.877	0.913
	3	0.040	0.026	0.041	12	0.961	0.939	0.959
	4	0.090	0.061	0.087	13	0.985	0.974	0.983
	5	0.170	0.122	0.166	14	0.996	0.990	0.994
	6	0.285	0.218	0.280	15	0.999	0.997	0.998
	7	0.424	0.348	0.425	16	1.000	0.999	1.000
	8	0.576	0.500	0.575				
Normal approximation*)				Normal approximation*)				
n=3	v	P(V≤v)	p'	p''	v	P(V≤v)	p'	p''
	0	0.000	0.000	0.000	19	0.626	0.587	0.629
	1	0.000	0.000	0.000	20	0.704	0.670	0.709
	2	0.000	0.000	0.001	21	0.774	0.745	0.776
	3	0.000	0.001	0.001	22	0.834	0.808	0.837
	4	0.002	0.001	0.002	23	0.883	0.862	0.885
	5	0.005	0.002	0.003	24	0.921	0.905	0.922
	6	0.005	0.004	0.006	25	0.949	0.937	0.950
	7	0.010	0.008	0.011	26	0.969	0.960	0.969
	8	0.018	0.015	0.019	27	0.982	0.976	0.981
	9	0.031	0.024	0.031	28	0.990	0.985	0.989
	10	0.051	0.040	0.051	29	0.995	0.992	0.994
	11	0.079	0.063	0.078	30	0.998	0.996	0.997
	12	0.117	0.094	0.115	31	0.999	0.998	0.998
	13	0.166	0.138	0.164	32	1.000	0.999	0.999
	14	0.226	0.192	0.224	33	1.000	0.999	1.000
	15	0.296	0.255	0.291	34	1.000	1.000	1.000
	16	0.374	0.330	0.371	35	1.000	1.000	1.000
	17	0.457	0.413	0.456	36	1.000	1.000	1.000
	18	0.543	0.500	0.544				

*) p' is without and p'' with a continuity correction.

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